Two bounds for the planar small-disorder LFPP constant

Abstract

We give two bounds for the planar small-disorder LFPP coefficient in the normalization of logarithmic covariance coefficient one and kinetic penalty $\frac{1}{2}\int |X'|^2$. Using the smooth white-in-time action identification from the supplied LFPP manuscript, we prove that $0.76283 < k_2 \leq 3\pi^{2/3}/2^{7/3}$. The lower bound comes from a path selected successively across frequency bands. A Gaussian state records previously observed cosine fluctuations and changes the amplitude of subsequent sine-driven displacements. Finite-band Gaussian identities account for both the reward from this dependence and its kinetic cost. The upper bound uses a scalar comparison with a high-component elastic manifold, the finite-graph Parisi formula, and an explicit concave trial profile. The LFPP-to-action identification and the cited finite-graph formula are external inputs; neither is reproved here.

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1 Statement and normalization

At small disorder, a path can reduce its random length by bending toward favorable field values. The coefficient k_2 measures the leading saving per logarithmic scale. This note bounds that coefficient by constructing one family of paths for the lower bound and comparing the optimum with a more favorable Gaussian model for the upper bound.

We first state the action convention, since a change in its noise or kinetic normalization changes the numerical constants. For a smooth positive-definite covariance C on $\mathbb{R}$, let b_C be a centered Gaussian field with covariance

$$\mathbb{E}[b_C(x,t)b_C(y,u)] = C(x-y)\delta(t-u)\,. \tag{1.1}$$

Write

$$\mathcal{K}(C) = \lim_{T\to\infty} \frac{1}{T}\mathbb{E} \sup_{X\in H_0^1(0,T)} \left\{ \int_0^T b_C(X(t),t)\,\mathrm{d}t - \frac{1}{2}\int_0^T |X'(t)|^2\,\mathrm{d}t \right\}. \tag{1.2}$$

The noise integral along a field-selected path is the pathwise integral defined by the Brownian primitive of b_C. It is not an adapted Itô integral in longitudinal time. If $B_C(x,t) = \int_0^t b_C(x,u)\,\mathrm{d}u$, the definition is

$$\int_0^T b_C(X(t),t)\,\mathrm{d}t = B_C(X(T),T) - B_C(X(0),0) - \int_0^T \partial_x B_C(X(t),t)X'(t)\,\mathrm{d}t\,. \tag{1.3}$$

For the smooth covariances used below, the existence and finiteness of (1.2), and its approximation by finite deterministic path families, are the inputs stated in [1, Proposition 3.2 and the paragraph following its proof]. The sign in (1.2) is equivalent, by Gaussian symmetry, to minus the expected minimum in that source.

Define

$$C_\varepsilon(x) = \sqrt{\frac{\pi}{2}} \int_{\varepsilon^2}^1 u^{-1/2} e^{-x^2/(2u)}\,\mathrm{d}u\,, \qquad 0<\varepsilon<1\,. \tag{1.4}$$

The specific LFPP input is

$$k_2 = \lim_{\varepsilon\downarrow 0} \frac{\mathcal{K}(C_\varepsilon)}{\log(1/\varepsilon)}\,. \tag{1.5}$$

This is [1, Theorem 4.1 and equation (4.2)]. Together with [1, Theorem 1.1], Theorem 4.1 and equation (4.2) identify this action limit with $\lim_{\xi\downarrow 0}\xi^{-4/3}\chi_2(\xi)$, where $\chi_2(\xi)$ is the planar LFPP distance exponent of [1], when the logarithmic field covariance has coefficient one and the LFPP weight has no Wick normalization. This note does not assume the two numerical bounds it will prove.

Theorem 1.1. *Under the action identification* (1.5),

$$L_* \le k_2 \le U_*\,, \tag{1.6}$$

where

$$L_* = \frac{18943\sqrt{2}}{19200} e^{-80189/256000} - \frac{53125}{248832} - \frac{279841}{6400000} > \frac{76283}{100000}\,, \tag{1.7}$$

$$U_* = \frac{3\pi^{2/3}}{2^{7/3}} = 1.27688322055378\ldots\,. \tag{1.8}$$

In fact, the lower argument bounds the lower limit in (1.5)*, and the upper argument bounds its upper limit, without using the existence of that limit.*

The lower proof retains information from previously observed frequency bands. A cosine-driven Gaussian state changes how strongly the next sine band moves the path. The sine reward can be calculated after conditioning on the entire cosine noise; the extra cosine reward is the derivative of that same conditional Gaussian characteristic function. To avoid an anticipative-integral shortcut, we calculate both rewards at a finite band partition. Only the resulting ordinary expectations are passed to the fine-partition limit.

The upper proof starts from a finite massive periodic chain. Convexity of its covariance compares a collection of independent scalar chains with a radial high-component chain. The Parisi formula computes the latter's limiting free energy. A concave trial profile bounds it at every radius. A resolvent comparison and a finite-family approximation then return to the scalar pinned action. The order of these limits is part of the proof.

The new lower construction is developed in Sections 2–4; Sections 5–6 give the upper proof. Appendix A records a rational check of the strict decimal threshold. The earlier manuscripts [2, 3] supply the two starting arguments; the finite-band implementation and the endpoint and limiting details are written out here.

2 Spectral bands and action scaling

The lower construction is easiest for an exact power-law spectral density on a finite frequency interval. We compare this field with (1.4) only after estimating its action. This uses an inequality between continuous spectral measures, not an identification of a frequency atom with a band.

The one-sided cosine convention is $C(x) = \int_0^\infty \rho(k)\cos(kx)\,\mathrm{d}k$. The Gaussian Fourier integral gives

$$\rho_\varepsilon(k) = \frac{2}{k^2}\left(e^{-\varepsilon^2 k^2/2} - e^{-k^2/2}\right). \tag{2.1}$$

Indeed, the cosine density of $e^{-x^2/(2u)}$ is $\sqrt{2u/\pi}\,e^{-uk^2/2}$; multiplication by $\sqrt{\pi/2}\,u^{-1/2}$ and integration in u gives (2.1).

For $S > 0$, put

$$C_S^{\mathrm{sh}}(x) = \int_1^{e^S} \frac{2}{k^2}\cos(kx)\,\mathrm{d}k\,. \tag{2.2}$$

Both this covariance and (1.4) have finite spectral moments of every order. If $C_2 - C_1$ is positive definite, adding an independent centered Gaussian field and applying conditional Jensen to each finite-family maximum gives $\mathcal{K}(C_2) \geq \mathcal{K}(C_1)$. The finite-family approximation in (1.2) passes this comparison to all paths.

Two exact scalings will be used:

$$\mathcal{K}(cC) = c^{2/3}\mathcal{K}(C)\,, \qquad \mathcal{K}(rC(\cdot/r)) = \mathcal{K}(C)\,, \qquad c, r > 0\,. \tag{2.3}$$

For the first, set $t = c^{-1/3}u$ in a finite-horizon action. White-time Gaussian scaling and the kinetic term both contribute the factor $c^{1/3}$, and division by the time horizon contributes another $c^{1/3}$. For the second, set $t = ru$ and $X(t) = rY(u)$. Both action terms and the horizon are multiplied by r. These identities also follow from [1, equation (3.8)].

Fix $R > 1$ and set $c_R = e^{-1/(2R^2)} - e^{-R^2/2}$. On $R \leq k \leq (R\varepsilon)^{-1}$, equation (2.1) is at least $2c_R/k^2$. With $S = \log(1/\varepsilon) - 2\log R > 0$, the covariance on that band is $R^{-1}C_S^{\mathrm{sh}}(Rx)$. Thus

$$\mathcal{K}(C_\varepsilon) \geq c_R^{2/3}\mathcal{K}(C_S^{\mathrm{sh}})\,. \tag{2.4}$$

Since $c_R \uparrow 1$, a lower bound for $\liminf_{S\to\infty} S^{-1}\mathcal{K}(C_S^{\mathrm{sh}})$ transfers to the lower limit in (1.5): first send $\varepsilon \downarrow 0$ at fixed R, then send $R \to \infty$.

3 A path chosen across frequency bands

At a fixed longitudinal time, the additional displacement supplied by a new band is centered Gaussian after conditioning on the cosine noise. Its variance depends on a state built from earlier cosine bands. This permits a reward calculation that keeps, rather than discards, the dependence of later motion on earlier fluctuations.

Fix $\lambda > 0$ and $b \in \mathbb{R}$. Initially let φ, ψ be real even smooth compactly supported functions on $\mathbb{R}$, with $\|\psi\|_2^2 = 2\lambda$ and $v := \|\varphi\|_2^2 > 0$. Write

$$A = \varphi(0)\,, \quad D = \psi(0)\,, \quad e_\varphi = \|\varphi'\|_2^2\,, \quad e_\psi = \|\psi'\|_2^2\,, \quad a(c) = 1 - bc\,. \tag{3.1}$$

Let $(C_r)_{r\geq 0}$ be a stationary Gaussian Ornstein–Uhlenbeck process with covariance $e^{-\lambda|r-u|}$. Define

$$F_L(c) = \mathbb{E}\left[\exp\left\{-\frac{v}{2}\int_0^L e^{-2r}a(C_r)^2\,\mathrm{d}r\right\}\middle| C_0 = c\right]\,, \tag{3.2}$$

$$F(c) = \lim_{L\to\infty} F_L(c)\,, \qquad M = \mathbb{E}F(Z)\,, \qquad z = \frac{\mathbb{E}[ZF(Z)]}{M}\,, \quad Z \sim N(0,1)\,. \tag{3.3}$$

The integral in the exponent is finite almost surely since its expectation is $(1+b^2)/2$. In particular, $0 < M \leq 1$.

Proposition 3.1. *With the preceding definitions,*

$$\liminf_{\varepsilon\downarrow 0} \frac{\mathcal{K}(C_\varepsilon)}{\log(1/\varepsilon)} \geq \sqrt{2}M\{A(1-bz) + Dz\} - \frac{e_\varphi}{2}(1+b^2) - \frac{vb^2 e_\psi}{4(\lambda+1)}\,. \tag{3.4}$$

The same bound holds for limits of these kernels in $H^1(\mathbb{R})$, with $v > 0$ and $\|\psi\|_2^2 = 2\lambda$ retained.

3.1 The finite-band construction

Fix $S > 0$ and a partition $\Pi:\ 0 = s_0 < s_1 < \cdots < s_J = S$. We will first obtain a bound valid at this finite partition. Let $\mathbb{T}_T$ be a temporal circle of circumference T. Take T greater than twice a common support radius of φ and ψ. Periodize the kernels

$$\varphi_s(t) = e^{-s/2}\varphi(e^s t)\,, \qquad \psi_s(t) = e^{s/2}\psi(e^s t)\,. \tag{3.5}$$

No two terms in either periodized sum are simultaneously nonzero. Thus their values at zero and their L^2 and derivative norms are exactly the real-line values:

$$\varphi_s(0) = e^{-s/2}A\,, \quad \psi_s(0) = e^{s/2}D\,, \quad \|\varphi_s\|_2^2 = e^{-2s}v\,, \quad \|\varphi_s'\|_2^2 = e_\varphi\,, \quad \|\psi_s'\|_2^2 = e^{2s}e_\psi\,. \tag{3.6}$$

All convolutions below are on this circle.

Start with two independent real white noises η^c, η^s on $[0,S]\times\mathbb{T}_T$, and an independent standard Gaussian Z_0. Set

$$C_s(t) = e^{-\lambda s}Z_0 + \int_0^s\int_{\mathbb{T}_T} e^{-\lambda(s-r)}\psi_r(t-u)\eta^c(\,\mathrm{d}r,\,\mathrm{d}u)\,. \tag{3.7}$$

For each fixed t, this is precisely the stationary OU process used in (3.2): the scale increment has variance $2\lambda\,\mathrm{d}s$ and the initial variance is one. The initial value is constant in t and therefore has no derivative cost.

For $0 \le i < J$, define

$$a_i(t) = a(C_{s_i}(t))\,, \quad Y_i(t) = \int_{s_i}^{s_{i+1}} \int_{\mathbb{T}_T} \varphi_r(t-u)\eta^s(\,\mathrm{d}r,\,\mathrm{d}u)\,, \quad X_i(t) = \sum_{\ell<i} a_\ell(t)Y_\ell(t)\,. \tag{3.8}$$

The chosen periodic path is X_J. The coefficient a_i and the reference path X_i use only bands strictly preceding band i. In particular, neither uses that band's sine noise.

We now construct a field of exactly the required law. For $s_i \le s < s_{i+1}$, put $\theta(s,t) = e^s X_i(t)$ and rotate the full pair of white noises:

$$\begin{pmatrix} W^c \\ W^s \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \eta^c \\ \eta^s \end{pmatrix}\,. \tag{3.9}$$

Equation (3.9) means multiplication of white noise by the displayed random orthogonal matrix on each band. Conditional on all earlier bands and Z_0, the matrix is fixed, so the new pair is a standard white noise pair with conditional law independent of that conditioning. More explicitly, for any deterministic pair of L^2 tests f_c, f_s supported in that band, its conditional characteristic functional is

$$\mathbb{E}\Big[\, e^{i(W^c(f_c)+W^s(f_s))} \Big|\, \mathcal{F}_{s_i}\Big] = \exp\{-\tfrac{1}{2}(\|f_c\|_2^2 + \|f_s\|_2^2)\}\,, \tag{3.10}$$

where $\mathcal{F}_{s_i}$ contains Z_0 and both noises in earlier bands. Conditional multiplication is defined first for simple earlier-measurable coefficients and then by its L^2 isometry. Iterating (3.10) proves that (W^c, W^s) is a full independent white noise pair on the entire rectangle, also independent of Z_0.

The spectral field built from this pair has action

$$\mathcal{H}_S(X) = \int_0^S \int_{\mathbb{T}_T} \sqrt{2}e^{-s/2}\{\cos(e^s X(t))W^c(\,\mathrm{d}s,\,\mathrm{d}t) + \sin(e^s X(t))W^s(\,\mathrm{d}s,\,\mathrm{d}t)\}\,. \tag{3.11}$$

For deterministic X, this has exactly the covariance of (2.2). For smooth selected X, the pairing agrees with (1.3). The constructed path is smooth in t almost surely. Substituting (3.9), its contribution on band i has factors

$$\cos(e^s(X_J - X_i))\,, \qquad \sin(e^s(X_J - X_i)) \tag{3.12}$$

in front of η^c and η^s, respectively. This cancellation of the earlier reference path is exact at a finite partition.

3.2 The expected kinetic cost

Conditional on the entire cosine noise, the Y_i are independent centered Gaussian functions. Also $\mathbb{E}[Y_i(t)Y_i'(t)] = 0$, since $\int \varphi_r\varphi_r' = 0$. Different i therefore have zero expected derivative cross terms, and

$$\mathbb{E}|(a_iY_i)'|^2 = (1+b^2)e_\varphi(s_{i+1}-s_i) + b^2 v I_i \mathbb{E}|\partial_t C_{s_i}|^2\,, \qquad I_i = \int_{s_i}^{s_{i+1}} e^{-2r}\,\mathrm{d}r\,. \tag{3.13}$$

Differentiating (3.7) and using (3.6) gives

$$\mathbb{E}|\partial_t C_s(t)|^2 = \frac{e_\psi}{2(\lambda+1)}(e^{2s} - e^{-2\lambda s})\,. \tag{3.14}$$

Since $e^{2s_i}I_i \le s_{i+1} - s_i$, the expected cost per unit temporal length satisfies

$$\frac{1}{2T}\mathbb{E}\int_{\mathbb{T}_T} |X_J'|^2\,\mathrm{d}t \le S\left\{\frac{e_\varphi}{2}(1+b^2) + \frac{vb^2e_\psi}{4(\lambda+1)}\right\}\,. \tag{3.15}$$

The second term is the additional cost of making the amplitude depend on the earlier cosine state. It cannot be omitted.

3.3 The exact finite-band reward

At one fixed time, abbreviate $a_i = a(C_{s_i})$ and put

$$V_i = v\sum_{\ell=i}^{J-1} a_\ell^2 I_\ell\,, \qquad T_i(s) = \sum_{\ell=i+1}^{J-1} e^{-\lambda(s_\ell - s)} a_\ell I_\ell\,, \quad s_i \le s < s_{i+1}\,. \tag{3.16}$$

The first quantity is the conditional variance of the remaining displacement $X_J - X_i$. The second records how a cosine increment in band i changes the amplitudes in later bands. All expectations in the next formula refer to the single stationary OU process (C_s).

Lemma 3.2. *The construction at partition* Π *satisfies*

$$\frac{1}{T}\mathbb{E}\mathcal{H}_S(X_J) = \sqrt{2}\sum_{i=0}^{J-1}\int_{s_i}^{s_{i+1}} \mathbb{E}\left[e^{-e^{2s}V_i/2}\{Aa_i + Dbve^{2s}T_i(s)\}\right]\,\mathrm{d}s\,. \tag{3.17}$$

The expression is independent of the circumference once the support condition above is met.

Proof. Condition on the entire cosine noise. The displacement $X_J(t) - X_i(t)$ is then centered Gaussian, of variance $V_i(t)$. Its derivative with respect to a sine-noise coordinate at (s,u) in band i is $a_i(t)\varphi_s(t-u)$. Gaussian integration by parts in that coordinate gives a diagonal contraction $e^s a_i(t)\varphi_s(0)$ times the expected cosine. Multiplication by the spectral factor $\sqrt{2}e^{-s/2}$ leaves $\sqrt{2}Aa_i(t)$. The conditional cosine expectation is $e^{-e^{2s}V_i(t)/2}$. This gives the first term in (3.17).

For the cosine contribution, first integrate over the sine noise. The coefficient of η^c becomes $e^{-e^{2s}V_i(t)/2}$. A cosine coordinate at (s,u) in band i has no effect on $a_i(t)$, whereas for $\ell > i$ it has derivative

$$D^c_{s,u} a_\ell(t) = -be^{-\lambda(s_\ell - s)}\psi_s(t-u)\,.$$

Consequently,

$$D^c_{s,u} V_i(t) = -2bv\psi_s(t-u)T_i(s,t)\,. \tag{3.18}$$

Differentiating the exponential, contracting at $u = t$, and using $\sqrt{2}e^{-s/2}\psi_s(0) = \sqrt{2}D$ gives the second term of (3.17). In particular, it is not legitimate to declare the future displacement independent of the cosine increment.

Here is a precise way to justify these contractions without replacing the rotated noise by a finite-dimensional noise. Keep the full white noises throughout (3.7)–(3.9). On each fixed band, expand the pairing of a noise with its random test function in a deterministic tensor basis, Fourier in t and a smooth spectral basis in s. The test functions in (3.12), and their Gaussian derivatives, have derivatives of every order within that band with finite moments. This follows from compact smooth kernels: C_{s_i} is Gaussian, and X_J is a finite sum of products of Gaussian smooth functions. The trigonometric factors and their derivatives have polynomial bounds in these functions.

Here are the summability and trace details. Use a cosine basis on the closed spectral interval and a Fourier basis on the temporal circle. Two integrations by parts in the spectral coordinate give $O((1+j)^{-2})$ coefficients, including their endpoint terms; arbitrarily many temporal integrations by parts give $O((1+|n|)^{-p})$ for any fixed p. Their random constants have finite second moments. Cauchy–Schwarz thus makes the series $\sum_{j,n}\mathbb{E}|\xi_{j,n}\langle g, e_{j,n}\rangle|$ finite. For each coefficient, conditioning on all other Gaussian coordinates and integrating the one remaining normal variable proves

$$\mathbb{E}[\xi_{j,n}\langle g, e_{j,n}\rangle] = \mathbb{E}\langle D_{e_{j,n}} g, e_{j,n}\rangle\,. \tag{3.19}$$

In the present application the kernel $D_{r,u}g(s,t)$ is smooth in the two spectral variables on the fixed band and smooth in the temporal variables, with the same moment bounds. Freezing at s_i is important here: the cosine derivatives use only $s_\ell > s$ with $\ell > i$, so there is no internal moving indicator on this band. Apply the preceding coefficient estimates in both variables. The double series of this kernel converges absolutely, and integration of its diagonal is the sum of its diagonal basis coefficients. Summing (3.19) therefore gives $\mathbb{E}W(g) = \mathbb{E}\int D_{s,t}g(s,t)\,\mathrm{d}s\,\mathrm{d}t$. Its diagonal values are the contractions computed above. This also proves convergence in L^1 of the truncated calculations. Distributional multiplication by the smooth temporal coefficients, or equivalently integration by parts with their Brownian primitives, identifies the pairings with (1.3). No assumption of adaptedness in t is made.

Finally, translation invariance of the periodized kernels makes the expectation independent of t. Integration over the circle contributes the factor T. □

3.4 Pinned paths and the limits

Cut the temporal circle at zero and add an interval of length one at each end. Join zero to $X_J(0)$ and $X_J(T) = X_J(0)$ to zero by straight segments. Give the added intervals fresh independent Gaussian fields with covariance (2.2). Conditional on the bulk construction, both connector rewards have mean zero. Their total kinetic cost is $X_J(0)^2$, and

$$\mathbb{E}X_J(0)^2 = (1+b^2)v\int_0^S e^{-2r}\,\mathrm{d}r \le \frac{(1+b^2)v}{2}\,. \tag{3.20}$$

The completed field has the correct white-time law on the enlarged interval, and the completed path is an admissible pinned competitor. The auxiliary randomness Z_0 does not enlarge the class in (1.2): for every realization of the field and Z_0, the deterministic supremum over all pinned H^1 paths dominates the chosen path's action. Dividing by $T+2$ and sending $T\to\infty$ gives

$$\mathcal{K}(C_S^{\mathrm{sh}}) \ge R_\Pi - SE_0\,, \qquad E_0 = \frac{e_\varphi}{2}(1+b^2) + \frac{vb^2e_\psi}{4(\lambda+1)}\,, \tag{3.21}$$

where R_Π is the right side of (3.17).

Now keep S fixed and let $|\Pi|\to 0$. At a fixed time the same OU process can be used for every partition. Its increments satisfy $\|C_s - C_u\|_{L^p} \le C_p|s-u|^{1/2}$ on $[0,S]$. Thus the sums in (3.16) converge in every finite L^p, uniformly in the integration variable, to

$$v\int_s^S e^{-2r}a(C_r)^2\,\mathrm{d}r\,, \qquad \int_s^S e^{-\lambda(r-s)}e^{-2r}a(C_r)\,\mathrm{d}r\,,$$

respectively. The missing or extra fraction of the current band has norm $O_S(|\Pi|)$. Since $|e^{-x}-e^{-y}| \le |x-y|$ for $x,y\ge 0$, Hölder's inequality and the same Gaussian moment bounds pass the products in (3.17) to the limit in L^1. Their integrated error is $O_S(|\Pi|^{1/2})$.

Conditional on $C_s = c$, stationarity and the Markov property identify the remaining exponential with $F_{S-s}(c)$. Direct differentiation under its expectation gives

$$F_L'(c) = vb\mathbb{E}_c\left[e^{-(v/2)\int_0^L e^{-2r}a(C_r)^2\,\mathrm{d}r}\int_0^L e^{-(\lambda+2)r}a(C_r)\,\mathrm{d}r\right]. \tag{3.22}$$

Indeed, changing C_0 changes C_r by the deterministic factor $e^{-\lambda r}$. Formula (3.21) therefore implies

$$\mathcal{K}(C_S^{\mathrm{sh}}) \ge \sqrt{2}\int_0^S \left\{A\mathbb{E}[a(Z)F_L(Z)] + D\mathbb{E}F_L'(Z)\right\}\,\mathrm{d}L - SE_0\,. \tag{3.23}$$

As $L \to \infty$, $F_L(Z) \to F(Z)$ in Gaussian L^1, also after multiplication by $1 + |Z|$. Formula (3.22) is dominated in Gaussian L^1 by an integrable multiple of $\int_0^\infty e^{-(\lambda+2)r}(1 + |C_r|)\,\mathrm{d}r$. It therefore gives convergence of F_L' in that space. Gaussian integration by parts in the single variable Z yields $\mathbb{E}F'(Z) = \mathbb{E}[ZF(Z)] = Mz$. Cesàro averaging in (3.23), followed by (2.4), proves (3.4) for compact smooth kernels.

For the last statement of Proposition 3.1, approximate the kernels by even compact smooth functions in $H^1(\mathbb{R})$ and normalize the second kernel to squared norm 2λ. In one dimension, H^1 convergence gives convergence of the values at zero. Thus $A, D, v, e_\varphi, e_\psi$ converge. The functional in (3.2) depends on the kernels only through v; dominated convergence gives continuity of M and $Mz = \mathbb{E}ZF(Z)$ in $v > 0$. Pass the already proved numerical inequalities to the limit. This proves the proposition without requiring a pathwise limit of the constructed white-noise action.

4 Evaluation of the Gaussian trial

The remaining random quantities in Proposition 3.1 concern a single OU process, not an unknown law of an optimal LFPP path. We bound its exponentially tilted mean and normalizing factor directly.

Set $\lambda = 2$, take $b > 0$, and write

$$w(r) = e^{-2r}\,, \qquad Y = \int_0^\infty w(r)C_r\,\mathrm{d}r\,, \qquad Q = \int_0^\infty w(r)C_r^2\,\mathrm{d}r\,. \tag{4.1}$$

The covariance is $R(r,u) = e^{-2|r-u|}$. Integration on the two sides of $u = r$ gives

$$I(r) := \mathbb{E}[C_rY] = (r + 1/4)e^{-2r}\,, \tag{4.2}$$

$$\operatorname{Var} Y = \frac{1}{8}\,, \qquad J := \int_0^\infty w(r)I(r)^2\,\mathrm{d}r = \frac{29}{864}\,, \qquad \int_0^\infty e^{-4r}I(r)\,\mathrm{d}r = \frac{5}{72}\,. \tag{4.3}$$

For example, the integral for J is that of $(r + 1/4)^2e^{-6r}$, so it follows from $\int_0^\infty r^j e^{-ar}\,\mathrm{d}r = j!/a^{j+1}$.

Lemma 4.1. *For $v, b > 0$, the quantities in (3.3) satisfy*

$$M \geq \exp\left\{-\frac{v(1+b^2)}{4} + \frac{v^2b^2}{16} - \frac{29v^3b^4}{1728}\right\}\,, \tag{4.4}$$

$$\frac{vb}{4} - \frac{5v^2b^3}{72} \leq z \leq \frac{vb}{4}\,. \tag{4.5}$$

Proof. Since $\int_0^\infty w = 1/2$, equation (3.2) gives

$$M = e^{-v/4}\mathbb{E}\exp\{vbY - vb^2Q/2\}\,. \tag{4.6}$$

Tilt first by the linear Gaussian functional vbY. Its normalizing constant is $e^{v^2b^2/16}$, and under the resulting probability measure the process has mean $vbI(r)$ and unchanged covariance R. In particular, its mean of Q is $1/2 + v^2b^2J$. Jensen's inequality for $e^{-vb^2Q/2}$ proves (4.4).

For the mean, let $\widehat{\mathbb{P}}$ have density proportional to $\exp(vbY - vb^2Q/2)$ and write $\mu(r) = \widehat{\mathbb{E}}C_r$. Gaussian integration by parts gives

$$\mu(r) = vbI(r) - vb^2\int_0^\infty R(r,u)w(u)\mu(u)\,\mathrm{d}u\,. \tag{4.7}$$

For precision, approximate the process by conditional expectations on increasing finite sets of Gaussian coordinates and truncate the integrals in time. The approximating Y and Q converge in finite L^p spaces: Gaussian moments and $\int w < \infty$ control their tails, and conditional Gaussian projections converge on each finite interval. Their linear exponential moments are bounded uniformly, while the negative quadratic factor is at most one. Hölder's inequality therefore gives uniform integrability for the weights and for the products entering Gaussian integration by parts. Convergence in probability then passes that finite-dimensional identity to (4.7). This is also the equation obtained by completing the Gaussian square.

We need positivity of the tilted mean; positive definiteness of a covariance operator alone would not imply pointwise positivity. Here the covariance has an explicit positive Green kernel. On the half-line, R is the inverse of

$$\mathcal{A} = \frac{-\partial_r^2 + 4}{4}, \qquad y'(0) = 2y(0). \tag{4.8}$$

Indeed its derivative has jump -4 at $r = u$, it decays at infinity, and it satisfies the displayed boundary condition. Equation (4.7) is therefore

$$(\mathcal{A} + vb^2 w)\mu = vbw. \tag{4.9}$$

Its quadratic form is

$$\frac{1}{4}\int_0^\infty |y'|^2 \,\mathrm{d}r + \int_0^\infty |y|^2 \,\mathrm{d}r + \frac{1}{2}|y(0)|^2 + vb^2 \int_0^\infty w|y|^2 \,\mathrm{d}r.$$

It is coercive on $H^1(0,\infty)$. The solution of (4.7) belongs to this space: μ is bounded by Gaussian exponential moment estimates, and applying the Green operator to $w\mu$ and to w gives the assertion. Testing (4.9) against the negative part μ_- gives a nonpositive left side equal to the negative of the preceding quadratic form at μ_-, while the right side is nonnegative. Thus $\mu \geq 0$.

Using this positivity in (4.7) yields $\mu(r) \leq vbI(r)$. At zero, use that upper estimate in its subtracted integral, together with (4.3), to obtain

$$\frac{vb}{4} - v^2 b^3 \int_0^\infty e^{-4u} I(u) \,\mathrm{d}u \leq \mu(0) \leq \frac{vb}{4}.$$

Conditioning on C_0 in (4.6) identifies $\mu(0) = \mathbb{E}[ZF(Z)]/M = z$, proving (4.5). □

We now choose the parameters in the original trial [2]:

$$v = \frac{6}{5}, \quad b = \frac{1}{4}, \quad A = \frac{5}{6}, \quad D = \frac{23}{10}, \quad \varphi(t) = \frac{5}{6}e^{-125|t|/216}, \quad \psi(t) = \frac{23}{10}e^{-529|t|/400}. \tag{4.10}$$

These even exponential kernels belong to $H^1(\mathbb{R})$ and are covered by the approximation clause of Proposition 3.1. For $f(t) = ae^{-q|t|}$, its squared L^2 norm is a^2/q and its squared derivative norm is $a^2 q$. Hence

$$\|\varphi\|_2^2 = \frac{6}{5}, \quad e_\varphi = \frac{3125}{7776}, \quad \|\psi\|_2^2 = 4, \quad e_\psi = \frac{279841}{40000}. \tag{4.11}$$

The last norm constraint gives $\lambda = 2$ exactly.

Substitution in Lemma 4.1 gives

$$M \geq e^{-80189/256000}, \qquad \frac{47}{640} \leq z \leq \frac{3}{40}. \tag{4.12}$$

Since the factors used below are positive,

$$A(1-bz)+Dz \geq \frac{5}{6}\frac{157}{160}+\frac{23}{10}\frac{47}{640}=\frac{18943}{19200}\,. \tag{4.13}$$

The two kinetic costs are

$$\frac{e_\varphi}{2}(1+b^2)=\frac{53125}{248832}\,, \qquad \frac{vb^2e_\psi}{4(\lambda+1)}=\frac{279841}{6400000}\,. \tag{4.14}$$

Equations (3.4) and (4.12)–(4.14) prove the lower bound by L_* in Theorem 1.1. Appendix A verifies its strict comparison with 0.76283 using rational numbers only.

5 An upper bound from the elastic manifold

For the upper bound, the lower construction is irrelevant: we need to bound the reward of every path. We first compare a scalar chain with a high-component radial chain, whose limiting free energy is given by a precise finite-graph Parisi formula. This section proves the scalar trial inequality used in the earlier manuscript [3], including its return to pinned continuous-time paths.

Write $C(x)=B(x^2)$. We will use the argument for $B=B_\varepsilon$ from (1.4), so B is an integral mixture of e^{-qx} against a finite positive measure supported on $[1/2,1/(2\varepsilon^2)]$. In particular B is completely monotone and has the analytic mixture representation required by [4, 5]. Let $\mathcal{K}_a(C)$ denote (1.2) with kinetic penalty $(a/2)\int|X'|^2$, for $a>0$.

Proposition 5.1. *Let $a>0$ and $B=B_\varepsilon$ for some $0<\varepsilon<1$. Suppose $H:[0,\infty)\to(0,\infty)$ is C^1, nondecreasing and concave, with $H(0)>0$ and $\ell_H:=\lim_{z\to\infty}H'(z)>0$. Then*

$$\mathcal{K}_a(C)\leq\frac{1}{8a}\int_0^\infty H(z)^{-2}\,\mathrm{d}z+2\int_0^\infty H(z)B''(2z)\,\mathrm{d}z\,. \tag{5.1}$$

5.1 The finite model and the source formula

Fix a cycle with $M\geq 3$ vertices, mesh $h>0$, and length $T=Mh$. For N-component configurations $u=(u_j)_{j=0}^{M-1}$, define

$$E_{a,\mu,h}(u)=\frac{a}{2h}\sum_j|u_{j+1}-u_j|^2+\frac{\mu h}{2}\sum_j|u_j|^2\,, \qquad \mu>0\,, \tag{5.2}$$

with cyclic indices. Let the independent site potentials have covariance

$$\mathbb{E}[V_{N,j}(u)V_{N,j}(v)]=NhB(|u-v|^2/N)\,. \tag{5.3}$$

Let $Z_{N,\beta}$ be the Lebesgue integral over all configurations of $\exp\{\beta(\sum_j V_{N,j}(u_j)-E_{a,\mu,h}(u))\}$.

Compare this radial field with the sum of N independent scalar site fields, whose covariance at one site is $h\sum_{i=1}^N B((u_i-v_i)^2)$. Convexity of B gives

$$NB\left(\frac{1}{N}\sum_i(u_i-v_i)^2\right)\leq\sum_i B((u_i-v_i)^2)\,. \tag{5.4}$$

The point variances agree. Gaussian interpolation for the log partition function therefore gives

$$\frac{1}{\beta T}\mathbb{E}\log Z_{1,\beta}\leq\frac{1}{\beta TN}\mathbb{E}\log Z_{N,\beta}\,. \tag{5.5}$$

For the sign, the interpolation derivative is the diagonal covariance change minus its two-replica average. Here the diagonal change is zero and the radial off-diagonal covariance is smaller. The separable partition function factorizes, giving the scalar left side of (5.5). First interpolate on product cubes, for which the separable partition function factorizes, and then exhaust the configuration space. Quadratic mass and the Gaussian envelope established below justify the exhaustion.

Let L_M be the nonnegative graph Laplacian of the cycle and put $\mathrm{tr} = M^{-1}\,\mathrm{Tr}$. Define

$$R_h(v) = \mathrm{tr}(vI + (a/h)L_M)^{-1}\,, \quad K_h = R_h^{-1}\,, \quad \Lambda_h(s) = sK_h(s) - \mathrm{tr}\log(K_h(s)I + (a/h)L_M)\,. \tag{5.6}$$

In particular $\Lambda_h' = K_h$. The notation K_h denotes an inverse matrix resolvent, not the action rate $\mathcal{K}$.

Here is the exact external formula used. For $q > 0$ and a probability measure ν on $[0, q]$ supported away from q, let

$$x(u) = \nu([0, u])\,, \qquad \Delta(u) = \int_u^q x(v)\,\mathrm{d}v\,, \tag{5.7}$$

and choose $q_* \in (\sup\operatorname{supp}\nu, q)$. Theorems 1.3 and 1.5 of [4, 5], respectively, give

$$\lim_{N\to\infty} \frac{1}{MN}\mathbb{E}\log Z_{N,\beta} = \sup_{q>0}\inf_{\nu} P_{\beta,q}(\nu)\,, \tag{5.8}$$

where

$$\begin{aligned} 2P_{\beta,q}(\nu) = {} & \log(2\pi/\beta) - \beta\mu h q + \Lambda_h(\beta(q - q_*)) \\ & + \beta\int_0^{q_*} K_h(\beta\Delta(u))\,\mathrm{d}u - 2\beta^2 h\int_0^q x(u)B'(2(q-u))\,\mathrm{d}u\,. \end{aligned} \tag{5.9}$$

This value is independent of q_*. The source assumptions hold at each fixed graph and fixed ε: the mixture defining B_ε is supported on bounded positive Gaussian rates. Our sign of the potential is immaterial by symmetry. Our nonnegative L_M is the negative of the graph Laplacian in the source, and its external linear field is zero.

To check the temperature factors, use equation (1.11) of [5] at inverse temperature one with parameters $\beta\mu h$, $\beta a/h$, and $\beta^2 hB$. If the elastic coefficient is multiplied by β, its inverse resolvent is $\beta K_h(\beta s)$ and its primitive is $\Lambda_h(\beta s) - \log\beta$. These substitutions give (5.9). As a further check, with $B = 0$ the point mass $\nu = \delta_0$ and optimization in q give

$$\frac{1}{2}\{\log(2\pi/\beta) - \mathrm{tr}\log(\mu h I + (a/h)L_M)\}\,, \tag{5.10}$$

exactly the normalized logarithm of the ordinary Gaussian integral for (5.2).

5.2 A trial at every radius

Fix $\beta > H'(0)$. Concavity and $H(0) > 0$ imply that $H(z)/z$ is strictly decreasing from infinity to ℓ_H, so there is a unique $z_\beta > 0$ with $H(z_\beta) = \beta z_\beta$. Put

$$H_\beta(z) = \begin{cases} \beta z, & 0 \le z \le z_\beta, \\ H(z), & z \ge z_\beta. \end{cases} \tag{5.11}$$

The function is concave, and $0 \le H'_\beta \le \beta$. For every radius q, choose the distribution function

$$x(u) = \beta^{-1} H'_\beta(q-u)\,, \qquad 0 \le u < q\,, \tag{5.12}$$

with right-continuous values at a jump. It is nondecreasing and equals one near q, so the measure is admissible. For $q \le z_\beta$ it is δ_0. Also $\beta\Delta(u) = H_\beta(q-u)$.

For $q > z_\beta$, reversing the variable in (5.9) and combining the portion where $H_\beta(z) = \beta z$ with Λ_h gives

$$\frac{P_{\beta,q}}{\beta h} = e_\beta - \frac{\mu q}{2} + \frac{1}{2h}\int_{z_\beta}^{q} K_h(H(z))\,\mathrm{d}z + \int_0^q H'_\beta(z)[-B'(2z)]\,\mathrm{d}z\,, \tag{5.13}$$

where $e_\beta = [\log(2\pi/\beta) + \Lambda_h(H(z_\beta))]/(2\beta h) \to 0$ as $\beta \to \infty$, at fixed graph and H.

The covariance term satisfies

$$\begin{aligned}\int_0^q H'_\beta(z)[-B'(2z)]\,\mathrm{d}z &= H_\beta(q)[-B'(2q)] + 2\int_0^q H_\beta(z)B''(2z)\,\mathrm{d}z \\ &\le 2\int_0^\infty H(z)B''(2z)\,\mathrm{d}z\,.\end{aligned} \tag{5.14}$$

Indeed $H_\beta \le H$, and the tail on the last line is at least $H(q)[-B'(2q)]$ by monotonicity of H. For $q \le z_\beta$, the point-mass formula and monotonicity of Λ_h give the same upper estimate. Thus (5.8) implies

$$\begin{aligned}\lim_{N\to\infty} \frac{\mathbb{E}\log Z_{N,\beta}}{\beta T N} \le e_\beta + 2\int_0^\infty H(z)B''(2z)\,\mathrm{d}z \\ + \sup_{q\ge 0}\left\{-\frac{\mu q}{2} + \frac{1}{2h}\int_0^q K_h(H(z))\,\mathrm{d}z\right\}\,.\end{aligned} \tag{5.15}$$

The trial is valid for every q before taking the supremum. No convergence of optimizing radii or Parisi measures is required.

5.3 The resolvent and the scalar limit

For a periodic affine interpolant f of the values u_j,

$$\int_0^T |f'|^2\,\mathrm{d}t = h^{-1}\sum_j |u_{j+1}-u_j|^2\,, \qquad \int_0^T f^2\,\mathrm{d}t = h\sum_j u_j^2 - \frac{h}{6}\sum_j (u_{j+1}-u_j)^2\,. \tag{5.16}$$

The variational characterization of point evaluation therefore bounds the discrete diagonal resolvent by its continuum counterpart:

$$\begin{aligned}R_h(hv) &\le \frac{1}{T}\sum_{k\in\mathbb{Z}} \frac{1}{v + a(2\pi k/T)^2} \\ &= \frac{1}{2\sqrt{av}}\coth\left(\frac{T}{2}\sqrt{v/a}\right) \le \frac{1}{2\sqrt{av}} + \frac{1}{Tv}\,.\end{aligned} \tag{5.17}$$

To see the first inequality, the continuum quadratic form at the interpolant is no larger than the discrete one by (5.16); taking the supremum of $|f(0)|^2$ divided by the form gives the claim. Cyclic symmetry identifies the diagonal inverse with the normalized trace. The Fourier series of the periodic Green function gives the displayed equality.

Invert the decreasing resolvent. Solving $s \le (2\sqrt{av})^{-1} + (Tv)^{-1}$ for $\sqrt{v}$ gives $\sqrt{v} \le (2\sqrt{a}\,s)^{-1} + (Ts)^{-1/2}$. Therefore, for every $\eta > 0$,

$$\frac{K_h(s)}{h} \le \frac{1+\eta}{4as^2} + \frac{1+\eta^{-1}}{Ts}\,. \tag{5.18}$$

Insert (5.18) into (5.15), combine with (5.5), and take $\beta \to \infty$ at fixed scalar graph. The scalar expected ground-state gain per temporal length is at most

$$\frac{1+\eta}{8a}\int_0^\infty H^{-2}\,\mathrm{d}z + 2\int_0^\infty HB''(2z)\,\mathrm{d}z + E_{T,\mu,\eta}(H)\,, \tag{5.19}$$

where

$$E_{T,\mu,\eta}(H) = \sup_{q\ge 0}\left\{-\frac{\mu q}{2} + \frac{1+\eta^{-1}}{2T}\int_0^q H(z)^{-1}\,\mathrm{d}z\right\}. \tag{5.20}$$

Only a finite-dimensional scalar zero-temperature limit is used here. To justify both expectation and exhaustion, let $V(u) = \sum_j V_{1,j}(u_j)$. Its supremum in absolute value on every unit cube has the same finite mean and a Gaussian tail, uniformly in the cube's location. Finiteness of that mean follows from the bounded derivative variances of the smooth field on the unit cube. For every $\delta > 0$, a union bound over $z \in \mathbb{Z}^M$ applied to these tails shows that

$$R_\delta = \sup_{z\in\mathbb{Z}^M}\left\{\sup_{u\in z+[0,1]^M}|V(u)| - \delta|z|^2\right\}_+$$

has finite expectation (in fact a Gaussian upper tail after adjusting constants). Choose δ small enough relative to μh. Then the Hamiltonian is bounded above by $R_\delta + C - \mu h|u|^2/4$. Integrating this envelope gives an integrable upper bound for $\beta^{-1}\log Z_{1,\beta}$, uniformly for $\beta \ge 1$. Jensen's inequality on a fixed cube of volume one gives the integrable lower bound $\int_{[0,1]^M}(V - E_{a,\mu,h})\,\mathrm{d}u$. For each realization, continuity and the quadratic envelope give the Laplace limit by a neighborhood of a maximizing point and a Gaussian tail bound on its complement. Dominated convergence now gives the expected limit. The same envelope, at each fixed finite N, justifies the exhausted interpolation. There is no exchange of $N \to \infty$ with $\beta \to \infty$.

Since $H(z) \ge H(0) + \ell_H z$, we have $\int_0^q H^{-1} \le \ell_H^{-1}\log(1 + \ell_H q/H(0))$. Taking $\mu = T^{-3}$ and maximizing the scalar logarithmic expression yields

$$E_{T,T^{-3},\eta}(H) = O_{H,\eta}\left(\frac{1+\log T}{T}\right). \tag{5.21}$$

The bound is uniform in the mesh. In detail, with $D_0 = (1+\eta^{-1})/(2T\ell_H)$ and $c_0 = \ell_H/H(0)$, the supremum is zero if $2D_0c_0 \le \mu$; otherwise it is at most $D_0\log(2D_0c_0/\mu)$.

5.4 Return to pinned paths

Fix the target stiffness $a > 0$, take $T > a^{-1}$, and let $a_T = a - T^{-1} > 0$. Sample a pinned H^1 path at the M periodic vertices. Jensen's inequality on each mesh interval and the bound $|f(t)|^2 \le T\int_0^T |f'|^2$ give

$$E_{a_T,T^{-3},h}(f(jh)) \le \frac{a_T + T^{-1}}{2}\int_0^T |f'|^2\,\mathrm{d}t = \frac{a}{2}\int_0^T |f'|^2\,\mathrm{d}t\,. \tag{5.22}$$

For a fixed finite family of deterministic paths, the covariance of the sampled site rewards is $h\sum_j C(f(jh)-g(jh))$. As $h = T/M \downarrow 0$, it converges to $\int_0^T C(f(t)-g(t))\,\mathrm{d}t$. Thus the corresponding finite Gaussian expected maxima, with the original fixed kinetic penalties, converge. By (5.22), they are bounded by (5.19) with a_T in place of a, uniformly in h.

Increase finite families containing zero to a countable dense family of pinned paths. Subtract the common centered reward at zero; the maxima are now nonnegative and monotone, so the action approximation and monotone convergence apply. This proves the same bound for the full pinned expected action at horizon T. Let $T \to \infty$ and use (5.21), then let $\eta \downarrow 0$. This proves Proposition 5.1.

6 Evaluation of the upper trial

The upper constant is the minimum of one elementary coefficient. The Parisi profile must nevertheless be admissible on the full half-line, including both cutoff regions.

For $B_\varepsilon(s) = C_\varepsilon(\sqrt{s})$,

$$B_\varepsilon''(s) = \frac{\sqrt{\pi/2}}{4}\int_{\varepsilon^2}^1 u^{-5/2}e^{-s/(2u)}\,\mathrm{d}u\,. \tag{6.1}$$

For $\alpha > 0$, set $H = \alpha f_\varepsilon$, where

$$f_\varepsilon(z) = \begin{cases} \sqrt{z+\varepsilon^2}, & 0 \le z \le 1,\\ (z+1+2\varepsilon^2)/(2\sqrt{1+\varepsilon^2}), & z \ge 1. \end{cases} \tag{6.2}$$

The function is positive, C^1, increasing and concave, and has positive limiting slope. It is therefore a permissible trial in Proposition 5.1. Direct integration gives

$$\int_0^\infty f_\varepsilon(z)^{-2}\,\mathrm{d}z = \log(1+\varepsilon^{-2}) + 2\,. \tag{6.3}$$

Also $f_\varepsilon(z) \le \sqrt{z} + \varepsilon + (z-1)_+/2$. For $z \le 1$ this is $\sqrt{z+\varepsilon^2} \le \sqrt{z}+\varepsilon$. For $z \ge 1$, start from $f_\varepsilon(1) = \sqrt{1+\varepsilon^2} \le 1+\varepsilon$ and use slope at most $1/2$ and $\sqrt{z} \ge 1$.

Tonelli's theorem, (6.1), and

$$\int_0^\infty \sqrt{z}\,e^{-z/u}\,\mathrm{d}z = \frac{\sqrt{\pi}}{2}u^{3/2}\,, \quad \int_0^\infty e^{-z/u}\,\mathrm{d}z = u\,, \quad \int_1^\infty (z-1)e^{-z/u}\,\mathrm{d}z = u^2e^{-1/u}$$

give

$$2\int_0^\infty f_\varepsilon(z)B_\varepsilon''(2z)\,\mathrm{d}z \le \frac{\pi}{2\sqrt{2}}\log(1/\varepsilon) + \frac{3}{2}\sqrt{\frac{\pi}{2}}\,. \tag{6.4}$$

For the remainder terms, use $\varepsilon\int_{\varepsilon^2}^1 u^{-3/2}\,\mathrm{d}u \le 2$ and $\int_{\varepsilon^2}^1 u^{-1/2}e^{-1/u}\,\mathrm{d}u \le 2$. The coefficient of the logarithm comes only from the $\sqrt{z}$ term.

Apply (5.1) with $a = 1$. Equations (6.3) and (6.4) show

$$\mathcal{K}(C_\varepsilon) \le \left(\frac{1}{4\alpha^2} + \frac{\pi\alpha}{2\sqrt{2}}\right)\log(1/\varepsilon) + \frac{2+\log 2}{8\alpha^2} + \frac{3\alpha}{2}\sqrt{\frac{\pi}{2}}\,. \tag{6.5}$$

The logarithmic coefficient is minimized at $\alpha^3 = \sqrt{2}/\pi$, where it equals $3\pi^{2/3}/2^{7/3}$. Divide (6.5) by $\log(1/\varepsilon)$ and send $\varepsilon \downarrow 0$. This proves the upper half of Theorem 1.1.

Completion of the proof of Theorem 1.1. Proposition 3.1 and the exact substitutions in Section 4 give the lower limit at least L_*. The preceding upper estimate gives the upper limit at most U_*. Equation (1.5) identifies their common limit with k_2, and Appendix A gives $L_* > 0.76283$.

For the lower argument, each temporal-length limit was taken at a fixed spectral cutoff and finite band partition. The partition was then refined at fixed cutoff, before the cutoff tended to infinity. The spectral comparison keeps R fixed until after the LFPP covariance cutoff is removed; $R \to \infty$ and approximation of the exponential trial kernels are last. For the upper argument, ε and the trial profile remain fixed through the high-component limit, the scalar zero-temperature limit, the temporal mesh limit, the long-time limit, and $\eta \downarrow 0$. Only then does $\varepsilon \downarrow 0$. These separate orders give the two inequalities without interchanging large-component, fine-cutoff, or zero-temperature limits. □

A An exact check of the lower threshold

The mathematical construction and the numerical comparison are separate. This appendix proves only the strict arithmetic inequality in (1.7).

Put $q = 80189/256000$ and $r = 141421356/10^8$. Integer multiplication gives $0 < r$ and $r^2 < 2$. Since $0 < q < 1$, the odd alternating Taylor sum satisfies

$$0 < \sum_{j=0}^{7} \frac{(-q)^j}{j!} < e^{-q}\,. \tag{A.1}$$

The entirely rational number

$$L_{\mathrm{rat}} = \frac{18943}{19200} r \sum_{j=0}^{7} \frac{(-q)^j}{j!} - \frac{53125}{248832} - \frac{279841}{6400000} \tag{A.2}$$

therefore satisfies $L_{\mathrm{rat}} < L_*$. Reduction to a common denominator gives

$$L_{\mathrm{rat}} - \frac{76283}{100000} = \frac{845753680301774451477405160189427427579791640070 3}{156889558346899736494080000000000000000000000000000000000} > 0\,. \tag{A.3}$$

The accompanying Python script checks these integer and rational identities, as well as the norm and OU-integral arithmetic used in Section 4. It uses no simulation, floating-point quadrature, or confidence interval. Its success does not by itself verify Proposition 3.1 or the external Parisi formula.

References

[1] *Small disorder in log-correlated first passage percolation.* Supplied 27-page manuscript, file `lfpp-small-parameter.pdf`, consulted 19 October 2026. Theorem 1.1; Section 3, Proposition 3.2, equation (3.8); and Section 4, Theorem 4.1 and equation (4.2). The author list is not supplied in the attached copy.

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[3] *Bounds and a large-dimensional limit for the small-disorder LFPP constant.* Earlier AI-assisted manuscript, October 2026; source `lfpp_large_dimension_rewritten.tex`, Sections 2–3. The scalar upper-bound route is reproduced here with its external input stated explicitly.

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